

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

April/May 2013

EE-306 ELECTRICAL MACHINES III

Date: 30.4.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Why do we need to carry out parallel operation of alternators? Explain the conditions needed to be satisfied in synchronization of alternator. [07]
- Q - 2 (a) State the disadvantages of synchronous impedance method of voltage regulation. Discuss the Poitier triangle method in details. [07]
- (b) Derive the emf equation of a synchronous generator. What are the assumptions made for the same? [04]
- (c) Define the cording of coil and distribution of coil along with vector diagrams, what is the effect of the same on emf generated? [03]

OR

- Q - 2 (a) Draw the equivalent circuit and phasor diagrams of synchronous generator for lagging power factor, leading power factor, and unity power factor and derive the equation of induce emf for each phasor diagrams. [06]
- (b) Differentiate between hydro generator and turbo generators. [05]
- (c) Why is armature winding kept stationary in synchronous machines. [03]
- Q - 3 (a) Draw and explain synchronous generator capability curve. State the information we get from this curve. [07]
- (b) Two three-phase alternators operate in parallel. The rating of one machine is 200 MW and that of the other is 400 MW. The droop characteristics of their governors are 4% and 5% respectively from no load to full load. Assuming that the governors are operating at 50 Hz at no load, how would a load of 600 MW be shared between them? What will be the system frequency at this load? Repeat the problem if both the governors have a droop of 4%. [07]

OR

- Q - 3 (a) Write a short note on short circuit ratio of Generators. [05]
- (b) A 3 phase Y-connected alternator is rated at 1600 KVA, 13500 V. The armature effective resistance and synchronous reactance are 1.5 ohm and 30 ohm respectively per phase. Calculate the % regulation for a load of 1280 KW at p.f.
i) 0.8 leading, ii) Unity [04]
- (c) Explain the Slip test to be conducted on Synchronous Machines. [05]

SECTION - II

Q - 4 (a) Select appropriate choices

[03]

1.) If the input to the prime mover of an alternator connected to infinite bus is kept constant but the excitation is changed, then the

- (A) Reactive component of the output is changed.
- (B) Active component of the output is changed.
- (C) Both active and reactive component changes.
- (D) Both active and reactive component remains constant.

2.) In an alternator if the armature reaction produces magnetization, demagnetization and cross magnetization of the main field then the load connected would be

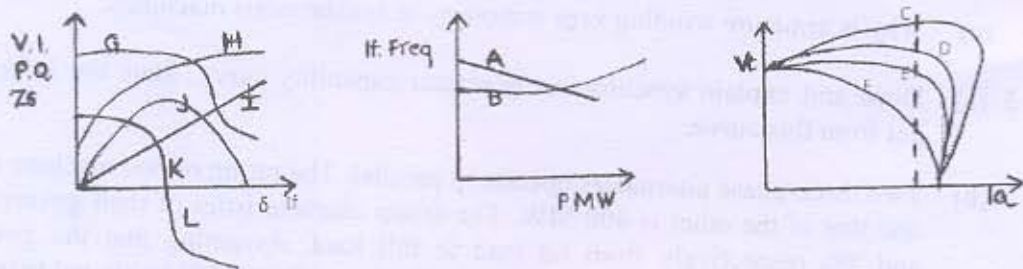
- (A) leading load, lagging load and unity power factor load
- (B) lagging load, leading load and unity power factor load
- (C) unity power factor load, leading load, and lagging load

3.) Alternators used in aircraft systems usually have frequency of

- (A) 25 Hz
- (B) 50 Hz
- (C) 100 Hz
- (D) 400 Hz.

(b) Identify (correlate) any 8 characteristic curve of synchronous machine from given options [04]

(P- δ , Load-Freq curve of SG, Compounding curve, 0.8 lagging-load characteristic, 0.9 leading-load characteristic, Unity power load characteristic, Q- δ curve, OCC curve, SC curve, Synchronous Impedance curve-- 0.8 leading-load characteristic)



Curve A =	Curve D =	Curve G =	Curve J =
Curve B =	Curve E =	Curve H =	Curve K =
Curve C =	Curve F =	Curve I =	Curve L =

Q - 5 (a) Explain the effect of varying the load of the synchronous motor and keeping the excitation constant, along with the necessary vector diagram. And explain the 'O' curve along with the conclusion derived from the same. [06]

(b) Draw and Explain the 'V' curve and inverted 'V' curve, of the synchronous machine. [05]

(c) Draw and explain the power flow diagram of synchronous motor. [03]

OR

- Q – 5 (a) Explain the effect of varying the field excitation while keep the load on synchronous motor constant. Also draw the necessary vector diagram. [06]
- (b) Explain the phenomenon of hunting or phase swinging in synchronous motors. [04]
- (c) Write a note on cooling systems of synchronous machines. [04]
- Q – 6 (a) Write a short note on Rosenberg generator. [07]
- (b) Explain the concept, construction, working and advantages of PMBLDC. [07]

OR

- Q - 6 (a) Write a short note on Induction Voltage Regulator. [05]
- (b) Derive the dynamic (swing) equation of synchronous machine. [05]
- (c) Write a short note on PMSM [04]
